

Progress Test 3.A — Thermo-S1

3.A · Low-density gases: the historical laws

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The product pV of a gas equals 2.0 bar L. Express it in joules.

[illegible]

Marking

①	
0	1
☐	☐

Exercise 2. Boyle–Mariotte law

(5 marks)

State the Boyle–Mariotte law: the conditions under which it holds, and its formula. Deduce from it the isothermal compressibility χ_T of a low-density gas.

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. The three historical laws

(3 marks)

Charles's law holds at fixed:

- ☐ n and T
☐ n and p
☐ n and V
☐ p and V

Gay-Lussac's law states that, at fixed n and V :

- ☐ $pV = \text{const.}$ ☐ $V/T = \text{const.}$ ☐ $p/T = \text{const.}$ ☐ $pT = \text{const.}$

For a low-density gas, the isobaric expansion coefficient is:

- ☐ $\alpha = 1/T$

☐ $\alpha = 1/p$

☐ $\alpha = T$

☐ $\alpha = 0$

Exercise 4. A rising bubble and a heated balloon

(5 marks)

(a) An air bubble of volume V_1 is released at a depth where the pressure is p_1 . It rises to the surface, where the pressure is p_2 , at constant temperature. Find its volume V_2 at the surface. (b) A balloon of volume V at θ_1 is heated to θ_2 at constant pressure. Find its new volume V' .

(a)	$V_1 = 2.0 \text{ cm}^3$	$p_1 = 3.0 \text{ bar}$	$p_2 = 1.0 \text{ bar}$
(b)	$V = 3.0 \text{ L}$	$\theta_1 = 27^\circ\text{C}$	$\theta_2 = 77^\circ\text{C}$

Marking

1

01

2

01

3

01

4

01

5

01

— End of the test —